

1(a)	(stress =) force / area or $\text{kg m s}^{-2} / \text{m}^2$	B1
	$= \text{kg m}^{-1} \text{s}^{-2}$	A1
1(b)(i)	$0.58 = 2\pi \times [(4 \times 0.500 \times 0.600^3) / (E \times 0.0300 \times 0.00500^3)]^{0.5}$	C1
	$E = [4\pi^2 \times 4 \times 0.500 \times (0.600)^3] / [(0.58)^2 \times 0.0300 \times (0.00500)^3]$ $= 1.35 \times 10^{10} \text{ (Pa)}$	C1
	$= 14 \text{ (13.5) GPa}$	A1
1(b)(ii)1.	(accuracy determined by) the closeness of the value(s)/measurement(s) to the true value	B1
	(precision determined by) the range of the values/measurements	B1
1(b)(ii)2.	l is (cubed so) $3 \times$ (percentage/fractional) uncertainty and T is (squared so) $2 \times$ (percentage / fractional) uncertainty and (so) l contributes more	B1